

EXPERIMENT 12

Image EXIF Metadata Examination

AIM

Extract available EXIF metadata from an image and organize camera, image, timestamp and GPS information.

ALGORITHM

1. Start.
2. Read/accept the required evidence data.
3. Process and analyze the data according to the experiment.
4. Apply the required threshold/rules/comparison.
5. Display the findings.
6. Stop.

CODE

```
from PIL import Image
from PIL.ExifTags import TAGS
img=Image.open("sample.jpg")
exif=img.getexif()
if not exif: print("No EXIF metadata available.")
else:
    for k,v in exif.items(): print(TAGS.get(k,k),":",v)
    if 34853 in exif: print("GPS information is available and may have forensic significance.")
```

OUTPUT

The screenshot shows a terminal window with a dark background. The top part displays the Python code being executed, which imports the Image module from PIL and the TAGS dictionary from PIL.ExifTags. It then opens a file named 'sample.jpg', retrieves its EXIF metadata, and prints it. The output shows the camera model 'Canon EOS 110D', the date and time '2020-03-21 07:30:23+05:30', and the GPS coordinates '30.123456, 76.123456'. The bottom part of the terminal shows the command prompt 'PS C:\Users\Adarsh\AppData\Local\Programs\Microsoft VS Code\...' and the file path 'C:\Users\Adarsh\AppData\Local\Programs\Microsoft VS Code\...'. The status bar at the bottom indicates the file is 'sample.jpg' and the Python version is 'Python 3.7.4'.

RESULT

The program extracts available EXIF metadata and continues even when fields are absent.